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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – May 2019

Course: B. Tech in Mechanical Engineering

Sem: IV

Subject Name: Manufacturing Processes I

Subject Code: BTMEC401

Max Marks: 60

Date: 14-05-2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		
A) What is meant by Manufacturing Processes? How will you select any suitable manufacturing process for the products?	K1 and K4/CO1	6
B) What is the requirement of shrinkage allowances? Explain the defects in the casting processes.	K1 and K2/CO1	6
C) Describe any suitable metal casting process for manufacturing the pump casing.	K6/CO1	6
Q.2 Solve the following.		
A) How will you roll any sheet material? Explain one rolling process in detail.	K3/CO2	6
B) Differentiate between open die and closed die forging with its suitability and applications	K4 and K5/CO2	6
Q.3 Solve Any One of the following.		
A) What do you understand by extrusion and drawing processes? Explain one of the both in detail.	K2 and K4/CO2	12
B) Define formability. Elaborate the sheet metal forming processing with various characteristics and types of shearing dies?	K1 and K2/CO2	12
Q.4 Write a short note on any Two of the following.		
A) Arc welding and Oxy-fuel gas welding processes	K1 and	6
B) Friction welding and resistance welding	K2/CO3	6
C) Soldering techniques and Mechanical fastening		6

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Q. 5 Solve Any One of the following.

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|---|----------------------|-----------|
| A) What are the types of lathe machines? Explain the various operations performed lathe machines with developments, attachment and accessories. | K3 and K4/CO4 | 12 |
| B) What is the need of drill machine and boring machines? Explain the various types of drilling machines with reaming and tapping processes | K3 and K4/CO4 | 12 |

Q. 6 Solve Any One of the following.

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|--|---------------|-----------|
| A) What is the need of milling machines? Comment on the face and end milling machines with its suitable applications and processes capabilities. | K6/CO5 | 12 |
| B) If you would like to fabricate gears for the gear box of any vehicle, which suitable manufacturing method will you select? Explain any two suitable gear manufacturing processes. | K6/CO6 | 12 |

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Supplementary Winter Semester Examination – Nov - 2019

Branch: Mechanical Engineering

Sem.:- IV

Subject :- Manufacturing Processes –I (BTMEC401)

Marks: 60

Date:- 26/11/2019

Time:- 3 Hr.

Instructions to the Students

1. Attempt **any five** questions of the following.
2. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
3. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

Q. N.	Question	Marks
1 a	Sketch the cross section of a sand mould which is ready for pouring and label the various important parts. Give a brief write-up on the following casting terms - Sprue, Gate, Runner, Riser.	4 2
1b	Explain the following properties of a moulding sand. i) Adhesiveness ii) Collapsibility iii) Permeability	6
2a	How does a cold rolling differ from hot rolling in terms of the process and product? Explain what do you understand by the terms slab and billet?	4 2
2b	Distinguish between open- and closed- die forging processes. What do you understand by the term flash in forging.	4 2
3a	A hole of 10 mm × 25 mm is to be cut in a 3 mm thick sheet. The shear strength of the material is 80 MPa. Estimate the press load required.	6
3b	Explain with sketches the difference between direct and indirect extrusion. List the variables which affect the extrusion process performance.	4 2
4a	List out any three differences between brazing and soldering. Write two applications of each of them. Explain the undercut and cracking defects in welding.	6
4b	Describe the submerged arc welding process in respect of working principle, advantages and limitations.	6
5a	List the methods of taper turning on a lathe. In a turning operation, a cutting speed of 55 m/min has been selected. At what rpm should a 15 mm diameter bar be rotated?	6
5b	Show with a neat sketch the various parts and angles of a twist drill. Explain the various allied operations that can be performed on drilling machine, draw a simple sketch.	6
6a	What are three basic forms of milling process? Differentiate between the up milling and down milling. Any three differences.	6
6b	Classify gear manufacturing methods. Explain gear hobbing process with a neat sketch.	6

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. ~~SE~~ . Branch : Mechanical Engineering

Semester ~~VI~~ IV

Subject Code & Name: BTMC401 Manufacturing Processes I

Max Marks: 60

Date: 13/08/2022

Duration: 3.45 Hr.

Instructions to the Students:

12/08/2022

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Answer Any Two of the following.

- A) Explain the following types of casting processes CO1 6
- I] Expendable-pattern casting (Lost Foam)
II] Shell-mold casting
- B) What is the function of chills in the casting processes? Why the chaplets are used in casting? CO1 6
- C) What is 'Fluidity Index'? Name the property of the mold that provides "ability to give way and allow casting to shrink without cracking the casting". Define the casting defect 'Misrun'. CO1 6

Q.2 Explain any two of the following with neat diagrams.

- A) Two-high or three-high rolling mills. CO2 6
- B) Cluster mill. CO2 6
- C) Impression-die forging. CO2 6

Q. 3 Answer Any Two of the following.

- A) Explain the Process of Impact Extrusion giving suitable example. CO3 6
- B) Explain types of shearing dies. CO3 6
- C) Show the following types of discontinuities in fusion welds. CO4 6
- i] Underfill
ii] Inclusions
iii] Overlap
iv] Undercut
v] Incomplete penetration

Q.4 Answer Any Two of the following.

- A) Explain the following types of welding processes CO4 6
- I] Thermit Welding
II] Friction Stir Welding
- B) What are different types of lathes? CO5 6
- C) What is gun drilling and how drill life is measured? CO5 6

Q. 5 Write short notes on Any Two of the following.

- A) Broaching. CO6 6
- B) End milling CO6 6
- C) Gear finishing processes. CO6 6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular End Semester Examination – Summer 2023 Course: B. Tech. Branch: Mechanical Engineering Semester: IV Subject Code & Name: BTMC401 /Manufacturing Processes – I Max Marks: 60 Date:13/07/2023 Duration: 3.00 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Define pattern? State any four types of patterns.	Remember	6
B)	What are the major types of sand moulds? What are their characteristics?	Understand	6
C)	Why is the investment-casting process capable of producing fine surface details on casting?	Understand	6
Q.2	Solve Any Two of the following.		
A)	Define roll gap, neutral point, and draft.	Remember/ Understand	6
B)	What is the difference between cold, warm, and hot forging.	Understand	6
C)	What is forgeability? Explain with neat diagram	Understand	6
Q. 3	Solve Any Two of the following.		
A)	What is SMAW? What are the advantages and applications of SMAW?	Understand	6
B)	What is the purpose of flux? Why is it not needed in gas tungsten-arc welding?	Remember	6
C)	Explain the principle involved in MIG welding?	Understand	6
Q.4	Solve Any Two of the following.		
A)	What are the components of a lathe?	Understand	6
B)	What do you mean by tool geometry? How single point turning tool geometry is specified?	Remember	6
C)	Find the time required to drill 4 holes in a cast iron flange each of 2 cm depth, if the	Analysis	6

	hole diameter is 2 cm. Assume cutting speed as 21.9 m/min. and feed as 0.02 cm/rev.		
Q. 5	Solve Any Two of the following.		
A)	Describe the different types of cutters used in milling operations and give an example of each one.	Understand	6
B)	Define broaching. Explain the geometric features of broach with their functions.	Remember	6
C)	Why do machined gears have to be subjected to finishing operations?	Remember	6
	*** End ***		

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter-2023 Course: B. Tech. Branch : MECHANICAL ENGINEERING/MECHANICAL ENGINEERING(SANDWICH) Semester : 4th Subject Code & Name: BTMC401 & Manufacturing Processes – I Max Marks: 60 Date: 16-01-24 Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Provide a schematic representation of the 'Mould Section and Casting Nomenclature' in sand casting, and articulate the functions associated with the terminologies used in sand casting	CO1	6
B)	Illustrate the schematic diagram depicting the Centrifugal Casting Process and elucidate its operational principles.	CO1	6
C)	Depict a schematic illustration of the Expandable Mold Casting Process and elaborate on its operational mechanisms	CO1	6
Q.2	Solve Any Two of the following.		12
A)	Categorize the metal forming techniques and provide a labeled sketch for each	CO2	6
B)	Classify the roll mills and draw the labelled sketches of each one of them.	CO2	6
C)	Categorize the metal forging processes, provide a sketch for each, and offer concise explanations for each process	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Illustrate and provide explanations for the following welding processes: 1. Gas Metal Arc Welding (GMAW).	CO3	6
B)	Sketch and explain the following Arc welding processes. 1. Flux Cored Arc Welding.	CO3	6
C)	Sketch and explain the following Arc welding processes. 2. Submerged Arc Welding.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Draw a labeled diagram illustrating the "Carriage" of a lathe machine, articulate its function, and enlist various components of it.	CO4	6

B)	Sketch and explain the construction and working of the “ Apron Mechanism ” of the Lathe machine.	CO4	6
C)	Illustrate and explain the construction and working of “ Tailstock Set-over Method ” of taper turning on lathe machine.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Draw and delineate different categories of "Peripheral Milling" operations, providing clear definitions for each type	CO5	6
B)	Illustrate and provide explanations for different variants of "Face Milling" operations.	CO5	6
C)	Define the term "Indexing," list various methods of indexing, and elucidate "Compound Indexing" with a relevant example	CO5	6
	*** End ***		

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Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define pattern. Explain any four types of patterns.	CO1	6
B) What is the function of core? What is core venting?	CO1	6
C) Explain the investment casting process with a sketch. Why is the investment casting process capable of producing fine surface details on castings?	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain any three examples of rolling stand arrangements.	CO2	6
B) Briefly explain about open die forging and closed die forging.	CO2	6
C) What are the different types of defects may occur in extrusion and drawing operations?	CO2	6
Q. 3 Solve Any Two of the following.		12
A) How cutting takes place when an oxyfuel gas torch is used? Also give its applications.	CO3	6
B) Explain the Gas Metal-arc Welding (MIG welding) with a neat sketch.	CO3	6
C) Describe the functions and characteristics of electrodes. What functions do coatings have?	CO4	6
Q.4 Solve Any Two of the following.		12
A) What are the different machining operations that can be performed on a lathe machine? Describe them with neat sketches.	CO3	6
B) Distinguish between the drilling and boring operations. Explain how can we perform both operations on a lathe?	CO4	6
C) What are the different types of chips generated in turning operations? Which type of chips are desirable?	CO4	6

Q. 5 Solve Any Two of the following.

12

- A)** What is the difference between end milling and face milling operations?
Explain them with neat sketches.
- B)** What do you mean by broaching operation? How it is done?
- C)** What are the different processes used for gear finishing? Explain them.

CO5

6

CO5

6

CO6

6

***** End *****